Additional case for GPK(Graham, Knuth and Patashnik) sequences

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Objective: To give the generating functions and study the asymptotic behavior of $T(n,k)$ defined by

$$T(n,k) = (a_2 n + a_1 k + a_0)\, T(n-1,k) + (b_2 n + b_1 k + b_0)\, T(n-1,k-1),$$

with the usual initial condition

$$T(0,0) = 1, \quad T(n,k) = 0 \text{ if } n < \max(k,0). \tag{1}$$

These sequences generalize many combinatorial numbers, such as Stirling numbers, Eulerian numbers, Lah numbers, and Whitney numbers. Graham, Knuth, and Patashnik posed this problem in their book [4]. Many years ago, Wilf [7] (and also Neuwirth [5]) found the generating function for the case $b_2 = 0$, which yields an explicit formula. Other cases remain open, both in terms of explicit formulas and asymptotics. For families generalizing Eulerian numbers, Hwang et al. [3] made a significant step using probabilistic methods to derive asymptotics in some cases as well. Hitczenko [8] did recently (2025) some cases using the generating function of $b_2 = 0$.

Here, we treat the case $a_2 = -a_1$ using context-grammar (Willian Chen sens) with Dumont tree-like structure associated (also system of differential equation combinatorial). We get the generating function and we try to give the asymptotic estimates and distribution of the random variables associated to $(T(n,k))$, using the basic theory in Flajolet [2].

I would like to emphasize the following This is not a piece of work intended for publication at this stage. It is simply part of my personal exercises on asymptotic methods that I studied earlier. Therefore, it has not undergone any formal verification, and I am not sure to what extent it contains new results or overlaps with existing work. It could potentially be combined with our paper, in preparation on context-free grammars (available on arXiv). This is a preliminary draft.

1 Generating functions for $a_1 + a_2 = 0$ (additional case)

We get these generating functions. The idea is to consider the grammar

$$\{x \to G(x) = (b_1+b_2)x^2+(a_1+a_2)xy; \quad y \to G(y) = b_2xy+a_2y^2; \quad z \to G(z) = (b_1+b_2+b_0)zx+(a_0+a_2)zy\} \tag{2}$$

and use the provable fact if $\mathcal{D} = G(x)\partial_x + G(y)\partial_y + G(z)\partial_z$,

$$\mathcal{D}^n(z) = z\sum_{k=0}^{n} T(n,k)x^k y^{n-k}. \tag{3}$$

We apply it in the Dumont [1] (Randrianirina [9]) point of views. What we use here, can be summarized as the following.

Proposition 1.1. *Let $\vec{y}(t) = (y_i(t))_{i=1,\dots,v}$ be the solution of the analytic differential system*

$$y_i'(t) = G_i(\vec{y}(t)), \qquad y_i(0) = x_i, \quad i = 1,\dots,v. \tag{4}$$

Let $\mathcal{D} = \sum_{i=1}^{v} G(\vec{x}(t))\partial_{x_i}$. Then for each i, $\mathrm{Gen}(x_i,t) := \sum_{n\geq 0} \mathcal{D}^n(x_i)\frac{t^n}{n!} = y_i(t)$.

In another hand, the data of the grammar G, the differential operator $\mathcal{D}$ and the differential system are equivalent. Here, we use only the analytic part of the differential system.
So firstly, we solve then the system of differential equation associated to the grammar (2).

$$\begin{cases} X'(t) &= (b_1+b_2)X^2, \quad X(0)=x, \\ Y'(t) &= b_2XY + a_2Y^2, \quad Y(0)=y, \\ Z'(t) &= (b_0+b_2)ZX + (a_0+a_2)ZY, \quad Z(0)=z \end{cases} \tag{5}$$

Applying proposition 1.1 with the fact (3), one can get

$$\sum_{n\geq 0} z \sum_{k=0}^{n} T(n,k)x^k y^{n-k}\frac{t^n}{n!} = \sum_{n\geq 0}\mathcal{D}^n(z)\frac{t^n}{n!} = Z(t,x,y,z).$$

We try to solve the system (5) and substitute $y=1$. So $F(t,x) = Z(t)/z = \sum_{n,k} T(n,k)x^k\frac{t^n}{n!}$.

....WILL WRITE THE CONCRETE (SHORT) WAY TO GET THE SOLUTION OF (5)....

1.1 Case 1: $b_1 + b_2 = 0$

Case 1A: $a_2 \neq 0$, $b_2 = 0$

$$F(t,x) = \sum_{n,k} T(n,k)x^k\frac{t^n}{n!} = \exp(b_0xt)\,(1-a_2t)^{-\frac{a_0+a_2}{a_2}}.$$

Case 1B: $a_2 = 0$, $b_2 = 0$

$$F(t,x) = \sum_{n,k} T(n,k)x^k\frac{t^n}{n!} = \exp(b_0xt + a_0t)\,.$$

Case 1C: $a_2 \neq 0$, $b_2 \neq 0$

$$F(t,x) = \sum_{n,k} T(n,k)x^k\frac{t^n}{n!} = \exp\big(b_0x\,t\big)\left(\frac{b_2x}{b_2x + a_2 - a_2\,e^{b_2xt}}\right)^{\frac{a_0+a_2}{a_2}}.$$

Then

$$T(n,k) = \binom{-\frac{a_0+a_2}{a_2}}{n-k}\left(\frac{a_2}{b_2}\right)^{n-k}\sum_{j=0}^{n-k}(-1)^j\binom{n-k}{j}(b_0+jb_2)^n.$$

Case 1D: $a_2 = 0$, $b_2 \neq 0$

$$F(t,x) = \sum_{n,k} T(n,k)x^k\frac{t^n}{n!} = \exp\left(b_0xt + \frac{a_0}{b_2x}\left(e^{b_2xt}-1\right)\right).$$

Then

$$T(n,k) = \frac{(a_0/b_2)^{n-k}}{(n-k)!}\sum_{j=0}^{n-k}(-1)^{n-k-j}\binom{n-k}{j}(b_0+jb_2)^n.$$

1.2 Case 2: $b_1 + b_2 \neq 0$, $a_2 = 0$

Case 2A: $b_1 \neq 0$

$$F(t,x) = \sum_{n,k} T(n,k)x^k\frac{t^n}{n!} = (1-(b_1+b_2)xt)^{-\frac{b_0+b_1+b_2}{b_1+b_2}}$$
$$\times \exp\left[-\frac{a_0}{b_1x}\left((1-(b_1+b_2)xt)^{\frac{b_1}{b_1+b_2}}-1\right)\right].$$

Case 2B: $b_1 = 0$

$$F(t,x) = \sum_{n,k} T(n,k) x^k \frac{t^n}{n!} = (1 - b_2 x t)^{-\frac{(b_0+b_2)x+a_0}{b_2 x}}.$$

Then

$$T(n,k) = a_0^{n-k} \sum_{1\le m_1<\cdots<m_k\le n} \prod_{r=1}^{k} (b_0 + m_r b_2)$$

Or

$$T(n,k) = a_0^{n-k}\; e_k\big(b_0+b_2, \ldots, b_0+nb_2\big)$$

1.3 Case 3: $b_1 + b_2 \neq 0$, $a_2 \neq 0$

Case 3A: $b_1 \neq 0$

$$F(t,x) = \sum_{n,k} T(n,k) x^k \frac{t^n}{n!} = (1-(b_1+b_2)xt)^{-\frac{b_0+b_1+b_2}{b_1+b_2}}$$

$$\times \left(\frac{1}{1 - \frac{a_2}{b_1 x}\left[1 - (1-(b_1+b_2)xt)^{\frac{b_1}{b_1+b_2}}\right]} \right)^{\frac{a_0+a_2}{a_2}}.$$

Case 3B: $b_1 = 0$

$$F(t,x) = \sum_{n,k} T(n,k) x^k \frac{t^n}{n!} = (1 - b_2 x t)^{-\frac{b_0+b_1}{b_2}-1}$$

$$\times \left(\frac{1}{1 + \frac{a_2}{b_2 x}\ln(1 - b_2 x t)} \right)^{\frac{a_0+a_2}{a_2}}.$$

2 Analytic approach

For this case we assume that $(T(n,k)) \subset \mathbb{R}_{\ge 0}$.

As usual, we set the random variable X_n defined by

$$\mathbb{P}(X_n = k) := \frac{T(n,k)}{\sum_{j=0}^{n} T(n,j)}.$$

Considering the polynom

$$P_n(x) := \sum_{k=0}^{n} T(n,k) x^k = n![t^n]F(t,x),$$

we have then $F(t,x) = \sum_{n\ge 0} P_n(x)\frac{t^n}{n!}$ and X_n is the random variable of probability generating function

$$p_n(x) = \mathbb{E}(x^{X_n}) = \frac{P_n(x)}{P_n(1)} = \frac{[t^n]F(t,x)}{[t^n]F(t,1)}.$$

2.1 Case 1A $a_1 = -a_2 \neq 0$, $b_1 = b_2 = 0$

$$F(t,x) = \sum_{n,k} T(n,k) x^k \frac{t^n}{n!} = \exp(b_0 x t)\,(1-a_2 t)^{-\frac{a_0+a_2}{a_2}}.$$

Proposition 2.1. *Let $(T(n,k))$ be sequence satisfying the usual initial condition (1) and*

$$T(n,k) = (a_2 n - a_2 k + a_0)\,T(n-1,k) + b_0\,T(n-1,k-1).$$

Assume $\frac{a_0+a_2}{a_2} \notin \mathbb{Z}_{\leq 0}$ and all coefficients are non-negative. $F(t,x)$ be the exponential bivariate generating function. Then The random variable X_n of probability generating function

$$p_n(x) = \mathbb{E}(x^{X_n}) = \frac{[t^n]F(t,x)}{[t^n]F(t,1)}$$

converges in distribution to a Poisson random variable.

Proof. We write the asymptotic of $p_n(x)$. First, we know

$$[t^n](1-a_2 t)^{-\alpha} = \frac{a_2^n}{\Gamma(\alpha)}\, n^{\alpha-1}\left(1 + O\left(\frac{1}{n}\right)\right).$$

Also $e^{b_0 x t}$ is analytic at the singularity $\rho = \frac{1}{a_2}$, then

$$[t^n]F(t,x) = e^{b_0 x/a_2} \frac{a_2^n}{\Gamma(\frac{a_0+a_2}{a_2})}\, n^{\frac{a_0}{a_2}}\left(1 + O\left(\frac{1}{n}\right)\right)$$

and

$$p_n(x) = \frac{e^{b_0 x/a_2}}{e^{b_0/a_2}} \cdot \frac{1+O(1/n)}{1+O(1/n)}.$$

So we have

$$p_n(x) \longrightarrow \exp(\lambda(x-1)), \qquad \lambda = \frac{b_0}{a_2}.$$

□

2.2 Case 1B $a_1 = a_2 = 0$, $b_1 = b_2 = 0$

$$F(t,x) = \sum_{n,k} T(n,k) x^k \frac{t^n}{n!} = \exp(b_0 x t + a_0 t).$$

This case is trivial, in many different ways, we can get quickly that

$$T(n,k) = \binom{n}{k} a_0^{n-k} b_0^k \qquad \Longrightarrow X_n \sim \mathrm{Bin}(n, \frac{b_0}{(a_0+b_0)}).$$

2.3 Case 1C: $a_1 = -a_2 \neq 0$, $b1 = -b_2 \neq 0$

$$F(t,x) = \sum_{n,k} T(n,k) x^k \frac{t^n}{n!} = \exp\big(b_0 x\, t\big)\left(\frac{b_2 x}{b_2 x + a_2 - a_2\, e^{b_2 x t}}\right)^{\frac{a_0+a_2}{a_2}}.$$

2.4 Case 1D: $a_2 = 0,\ b - 1 = -b_2 \neq 0$

$$T(n,k) = a_0\, T(n-1,k) + (b_2 n - b_2 k + b_0)\, T(n-1,k-1).$$

$$F(t,x) = \sum_{n,k} T(n,k) x^k \frac{t^n}{n!} = \exp\left(b_0 x t + \frac{a_0}{b_2 x}\left(e^{b_2 x t} - 1\right)\right). \tag{6}$$

This leads to

$$T(n,k) = \left(\frac{a_0}{b_2}\right)^{n-k} \sum_{r=n-k}^{n} \binom{n}{r} b_0^{n-r} b_2^r S(r, n-k),$$

$S(r,m)$ is Stirling numbers of the second kind.

But, the equation (6) is equal to

$$\begin{aligned}
\sum_{n,k} T(n,k)(\frac{1}{x})^{n-k} \frac{(xt)^n}{n!} &= \exp\left(b_0 x t + \frac{a_0}{b_2 x}\left(e^{b_2 x t} - 1\right)\right) \\
\implies \sum_{n,k} T(n,k) u^{n-k} \frac{z^n}{n!} &= e^{b_0 z} \exp\left(\frac{a_0}{b_2}\left(e^{b_2 x t} - 1\right) u\right) \\
\implies \sum_{n \geq 0} T(n,k) \frac{z^n}{n!} &= \frac{a_0^{n-k}}{(n-k)! b_2^{n-k}} e^{b_0 z} \left(e^{b_2 z} - 1\right)^{n-k}
\end{aligned}$$

From the Cauchy Integral formula, we get

$$T(n,k) = \frac{n!}{2\pi i} \frac{a_0^{n-k}}{(n-k)! b_2^{n-k}} \int_{\mathcal{C}(0,\rho)} \frac{e^{b_0 z}\left(e^{b_2 z} - 1\right)^{n-k}}{z^{n+1}}\, dz. \tag{7}$$

We can apply the Saddle point bound directly but the bounds is very high for some value that we try via Maple. One can do is, use similar thing what Corcino [10] estimate the integral, for $r - Whitney$ numbers.

Theorem 2.1. *Let* $(T(n,k))$ *be sequence satisfying the usual initial condition* (1) *and*

$$T(n,k) = a_0\, T(n-1,k) + (b_2 n - b2k + b_0)\, T(n-1,k-1).$$

Let $\rho^* > 0$ *be the unique solution of the saddle-point equation*

$$\rho^* \left(b_0 + \frac{(n-k) b_2}{1 - e^{-b_2 \rho^*}} \right) = n. \tag{8}$$

Define

$$B = \frac{1}{2} \left[\frac{b_0}{n-k} + \frac{b_2 e^{b_2 \rho^*}(e^{b_2 \rho^*} - b_2 \rho^* - 1)}{(e^{b_2 \rho^*} - 1)^2} \right]. \tag{9}$$

Then, for $\frac{b_0}{b2} \geq 4/3$,

$$T(n,k) \sim \frac{n!}{(n-k)!} \frac{a_0^{n-k}}{b_2^{n-k}} \frac{e^{b_0 \rho^*}(e^{b_2 \rho^*} - 1)^{n-k}}{2\,(\rho^*)^n \sqrt{\pi (n-k) \rho^* B}} (1+s), \tag{10}$$

with

$$
\begin{aligned}
s = \frac{1}{(n-k)\,\rho^*}\Bigg\{ & \frac{3}{4\rho^* B^2}\Bigg[\frac{b_0\rho^*}{24(n-k)} + \frac{1}{24}\Big(b_2\rho^* + (b_2\rho^* - 7(b_2\rho^*)^2 + 6(b_2\rho^*)^3 - (b_2\rho^*)^4)(e^{b_2\rho^*}-1)^{-1}\Big) \\
& + \frac{1}{24}\Big((18(b_2\rho^*)^3 - 7(b_2\rho^*)^2 - 7(b_2\rho^*)^4)(e^{b_2\rho^*}-1)^{-2}\Big) \\
& + (12(b_2\rho^*)^3 - 12(b_2\rho^*)^4)(e^{b_2\rho^*}-1)^{-3} - \frac{1}{24}\Big(6(b_2\rho^*)^4(e^{b_2\rho^*}-1)^{-4}\Big)\Bigg] \\
& - \frac{15}{16(\rho^*)^2B^3}\Bigg[\frac{b_0\rho^*}{6(n-k)} + \frac{1}{6}\Big(b_2\rho^* + (b_2\rho^* - 3(b_2\rho^*)^2 + (b_2\rho^*)^3)(e^{b_2\rho^*}-1)^{-1}\Big) \\
& + (3(b_2\rho^*)^3 - 3(b_2\rho^*)^2)(e^{b_2\rho^*}-1)^{-2} + \frac{1}{6}\Big(2(b_2\rho^*)^3(e^{b_2\rho^*}-1)^{-3}\Big)\Bigg]^2\Bigg\}.
\end{aligned}
$$

Also, it is still true even $q := \frac{b_0}{b2} < 4/3$, if $k \le n/q$.

Proof. From (7),

$$T(n,k) = \frac{n!}{2\pi i}\frac{a_0^{n-k}}{(n-k)!b_2^{n-k}}I, \quad I = \int_{\mathcal{C}(0,R)} \frac{e^{b_0 z}\big(e^{b_2 z}-1\big)^{n-k}}{z^{n+1}}\,dz. \tag{11}$$

We parametrize the contour by

$$z = \rho e^{i\theta}, \qquad dz = i\rho e^{i\theta}d\theta.$$

Substituting into the integral gives

$$I = \int_{-\pi}^{\pi} \frac{e^{b_0\rho e^{i\theta}}\big(e^{b_2\rho e^{i\theta}}-1\big)^{n-k}}{(\rho e^{i\theta})^{n+1}}\, i\rho e^{i\theta}\,d\theta = i\rho^{-n}\int_{-\pi}^{\pi}\exp\big(b_0\rho e^{i\theta}\big)\big(e^{b_2\rho e^{i\theta}}-1\big)^{n-k}e^{-in\theta}\,d\theta.$$

Thus,

$$I = i\rho^{-n}\int_{-\pi}^{\pi}\exp(h_\rho(\theta))\,d\theta, \quad h_\rho(\theta) = b_0\rho e^{i\theta} + (n-k)\,\log\big(e^{b_2\rho e^{i\theta}}-1\big) - in\theta.$$

The next step is expand $h_\rho(\theta)$ and choose ρ^*, saddle point (at $\theta = 0$): we choose is as radius of the countour. So in the expansion, the order zero is constant (can put out of the integral); the order is zero by definition.We have then

$$T(n,k) = (\textbf{constant}) \times \int_{-\pi}^{\pi}\exp\big(-(n-k)B\theta^2\big)\exp\big(\textbf{oder superior to 3}\big).$$

Now, we want to bring it to gamma function, so we need $x^2 = (n-k)B\theta^2$.
The next task is to argue one the domain of integral, so it can extend to infinity and many computation to have the result.

.... to complete the full writing...

Surely, there are conditions... □

$$F(t,x) = \sum_{n,k} T(n,k)x^k\frac{t^n}{n!} = \exp\left(b_0xt + \frac{a_0}{b_2x}\big(e^{b_2xt}-1\big)\right). \tag{12}$$

We can also see that $F_x(t) = F(t, x)$ is H-admissible for $x > 0$ with

$$\rho = +\infty, \quad R_0 > 0, \quad a(r) = r\left(b_0 x + a_0 e^{b_2 x r}\right),$$

$$b(r) = a_0 b_2 x\, r^2 e^{b_2 x r} + a_0 r e^{b_2 x r} + b_0 x\, r, \quad \theta_0(r) = r^{-\frac{1}{2}}.$$

Then Coefficients of admissible functions theorem ensures us

$$[t^n]F(t,x) \sim \frac{\exp\Big(b_0 x \zeta_x(n) + \frac{a_0}{b_2 x}\left(e^{b_2 x \zeta_x(n)} - 1\right)\Big)}{\zeta_x(n)^n \sqrt{2\pi\left(a_0 b_2 x\, \zeta_x(n)^2 e^{b_2 x \zeta_x(n)} + a_0 \zeta_x(n) e^{b_2 x \zeta_x(n)} + b_0 x \zeta_x(n)\right)}} \tag{13}$$

where $\zeta_x(n)\left(b_0 x + a_0 e^{b_2 x \zeta_x(n)}\right) = n$.

We want to try to use the Quasi-Powers, general distributions: Laplace transform

$$\lambda_n(s) = \mathbb{E}(e^{sX_n}) = \frac{[t^n]F(t, e^s)}{[t^n]F(t,1)},$$

is of form

$$\lambda_n(s) = \exp\big(\beta_n U(s) + V(s)\big)\left(1 + O\left(\frac{1}{\kappa_n}\right)\right),$$

with $\beta_n, \kappa_n \to +\infty$ as $n \to +\infty$, and $U''(0) \neq 0$. $U(s)$ and $V(s)$ is analytic near 0.

I tried but I don't see so far... after many try...

Now, let's try the Generalized quasi-Powers theorem. We need to write $p_n(x) = \exp(h_n(t,x))$. We need to start by approximating $p_n(x)$ near $x = 1$. If $h(t,x) = b_0 x t + \frac{a_0}{b_2 x}\left(e^{b_2 x t} - 1\right)$ then $F(t,x) = \exp(h(t,x))$ and the saddle point equation is

$$\frac{\partial}{\partial t} h(t,x)\Big|_{t=\zeta_x(n)} = \frac{n}{\zeta_x(n)} \implies \zeta_x(n)\Big(b_0 x + a_0 e^{x\zeta_x(n)}\Big) = n. \tag{14}$$

We get now (13) and compute

$$p_n(x) \sim \frac{F(\zeta_x(n), x)}{F(\zeta_1(n), 1)}\left(\frac{\zeta_1(n)}{\zeta_x(n)}\right)^n K(x)(1 + o(1)), \quad K(x) = \sqrt{\frac{n + a_0 b_2 \zeta_1(n)^2 e^{b_2 \zeta_1(n)}}{n + a_0 b_2 x \zeta_x(n)^2 e^{b_2 x \zeta_x(n)}}}.$$

The $h_n(x)$ wanted is

$$h_n(x) = \log p_n(x) = h(\zeta_x(n), x) - h(\zeta_1(n), 1) + n \log\left(\frac{\zeta_1(n)}{\zeta_x(n)}\right) + \log K(x) + o(1).$$

Checking carefully, on has

$$\frac{h_n'''(1)}{\left(h_n'(1) + h_n''(1)\right)^{3/2}} \to 0 \quad (n \to \infty).$$

Now, from the Generalized quasi-Powers theorem, we can conclude that the associated X_n converges in distribution to a normal law with mean $h_n'(1)$ and variance $h_n'(1) + h_n''(1)$.
using elementary calculus with equation (14) $h_n'(x) = \partial_x h(\zeta_n(n), x) + K'(x)/K(x)$. Then

$$\mu_n \sim h_n'(1) \sim b_0 \zeta_1(n) + a_0 \zeta_1(n) e^{b_2 \zeta_1(n)} - \frac{a_0}{b_2}(e^{b_2 \zeta_1(n)} - 1) + K'(x)/K(x). \tag{15}$$

If $D(x) = n + a_0 b_2 x \zeta_x(n)^2 e^{b_2 x \zeta_x(n)}$, $K'(x)/K(x) = -\frac{1}{2}\frac{D'(x)}{D(x)}$. Then

$$\frac{K'(x)}{K(x)} = -\frac{1}{2}\frac{a_0 b_2 e^{b_2 x \zeta_x(n)}\Big(\zeta_x(n)^2 + 2x\, \zeta_x(n)\, \partial_x \zeta_x(n) + b_2 x\, \zeta_x(n)^3 + b_2 x^2\, \zeta_x(n)^2\, \partial_x \zeta_x(n)\Big)}{n + a_0 b_2 x\, \zeta_x(n)^2 e^{b_2 x \zeta_x(n)}}, \tag{16}$$

with

$$\partial_x \zeta_x(n) = -\frac{b_0\,\zeta_x(n) + a_0 b_2\,\zeta_x(n)^2 e^{b_2 x \zeta_x(n)}}{b_0 x + a_0 e^{b_2 x\zeta_x(n)} + a_0 b_2 x\,\zeta_x(n) e^{b_2 x \zeta_x(n)}}. \tag{17}$$

So,

$$\frac{K'(x)}{K(x)} = -\frac{1}{2}\frac{a_0 b_2 e^{b_2 x\zeta_x(n)}}{n + a_0 b_2 x\,\zeta_x(n)^2 e^{b_2 x \zeta_x(n)}}\Bigg[\zeta_x(n)^2 + b_2 x\,\zeta_x(n)^3 \tag{18}$$

$$- \left(2x\,\zeta_x(n) + b_2 x^2\,\zeta_x(n)^2\right)\frac{b_0\,\zeta_x(n) + a_0 b_2\,\zeta_x(n)^2 e^{b_2 x\zeta_x(n)}}{b_0 x + a_0 e^{b_2 x \zeta_x(n)} + a_0 b_2 x\,\zeta_x(n) e^{b_2 x\zeta_x(n)}}\Bigg]. \tag{19}$$

As asymptotically, we have $b_0 x + a_0 e^{b_2\zeta_x(n)} \sim a_0 e^{b_2\zeta_x(n)}$, the saddle point equation gives us

$$\zeta_x(n) a_0 e^{b_2 \zeta_x(n)} \sim n. \tag{20}$$

So the prefactor is $\frac{a_0 b_2 e^{b_2 x\zeta}}{n + a_0 b_2 x \zeta^2 e^{b_2 x\zeta}} \sim \frac{1}{1 + x\zeta_x(n)} \sim \frac{1}{x\zeta_x(n)}$. The first part in bracket is $\zeta^2 + b_2 x\zeta^3 = \zeta^2(1 + b_2 x\zeta), \sim b_2 x \zeta^3$. For the fraction in bracket, exponential terms dominate: $\frac{b_0\zeta + a_0 b_2 \zeta^2 e^{b_2 x\zeta}}{b_0 x + a_0 e^{b_2 x\zeta} + a_0 b_2 x\zeta e^{b_2 x\zeta}} \sim \frac{\zeta}{x}$.

Thus the correction becomes $(2x\zeta + b_2 x^2\zeta^2)\frac{\zeta}{x} = 2\zeta^2 + b_2 x\zeta^3$.

Now we combine terms inside the bracket: $\zeta^2 + b_2 x\zeta^3 - (2\zeta^2 + b_2 x\zeta^3) = -\zeta^2$.

Finally, $\frac{K'(x)}{K(x)} \sim -\frac{1}{2}\cdot\frac{1}{x\zeta^2}\cdot(-\zeta^2) = \frac{1}{2x}$.

Therefore, $\frac{K'(x)}{K(x)} \sim \frac{1}{2x}$.

From (14), as $n \to \infty$, $\zeta(1) a_0 e^{\zeta(1)} \sim n$. Then $\log(\zeta) + \log(a_0) + \zeta \sim \log n$. Taking only the last part ($\zeta \sim \log(n)$) and plugin again, we have $\log(\log n) + \log a_0 + \zeta \sim \log n$. So

$$\zeta \sim \log n - \log(\log n) - \log(a_0) \implies \frac{a_0}{b_2} e^{b_2\zeta} \sim \frac{a_0^{1-b_2}}{b_2}\frac{n^{b_2}}{(\log n)^{b_2}}.$$

$$\frac{a_0}{b_2} e^{b_2\zeta} \sim \frac{a_0^{1-b_2}}{b_2}\frac{n^{b_2}}{(\log n)^{b_2}}.$$

I tried using computer tools for some values of a_0, b_0, b_2, and see

$$\mathbb{E}\{X_n\} \sim n - \frac{n}{\log n}, \qquad \mathbb{V}\{X_n\} \sim \frac{n}{(\log n)^2}.$$

.....ALL OF THE COMPUTATION AND CONDITIONS NEED TO CHECK AGAIN...

2.5 Case2A: $b_1 + b_2 \neq 0$, $a_2 = 0$ and $b_1 \neq 0$ ($b_1 < 0$)

$$F(t,x) = \sum_{n,k} T(n,k) x^k \frac{t^n}{n!} = (1 - (b_1+b_2)xt)^{-\frac{b_0+b_1+b_2}{b_1+b_2}}$$

$$\times \exp\left[-\frac{a_0}{b_1 x}\left((1-(b_1+b_2)xt)^{\frac{b_1}{b_1+b_2}} - 1\right)\right].$$

Note that the singularity at $x = 0$ is removable.

It is convenient to write $F(t,x) = \exp(h(t,x))$, where

$$h(t,x) = -\frac{b_0+b_1+b_2}{b_1+b_2}\log\big(1-(b_1+b_2)xt\big) - \frac{a_0}{b_1 x}\Big((1-(b_1+b_2)xt)^{\frac{b_1}{b_1+b_2}} - 1\Big).$$

The saddle point $\zeta_x(n)$ is defined implicitly by

$$\frac{\partial}{\partial t} h(t,x)\Big|_{t=\zeta_x(n)} = \frac{n}{\zeta_x(n)}.$$

So

$$\frac{(b_0+b_1+b_2)x}{1-(b_1+b_2)x\zeta_x(n)} + a_0\left(1-(b_1+b_2)x\zeta_x(n)\right)^{-\frac{b_2}{b_1+b_2}} = \frac{n}{\zeta_x(n)}.$$

We know that the saddle point approach to dominant singularity as $n \to \infty$, let's rewrite the above equation as

$$\frac{1}{\left(1-(b_1+b_2)x\zeta_x(n)\right)^{\frac{b_2}{b_1+b_2}}}\left(\left(1-(b_1+b_2)x\zeta_x(n)\right)^{-\frac{b_1}{b_1+b_2}} + a_0\right) = \frac{n}{\zeta_x(n)}.$$

So asymptotically,

$$1-(b_1+b_2)x\,\zeta_x(n) \sim \left(\frac{a_0}{(b_1+b_2)x\,n}\right)^{\frac{b_1+b_2}{b_2}}.$$

So

$$\zeta_x(n) = \frac{1}{(b_1+b_2)x}\left(1-\left(\frac{a_0}{(b_1+b_2)x\,n}\right)^{\frac{b_1+b_2}{b_2}} + o\left(n^{-\frac{b_1+b_2}{b_2}}\right)\right).$$

We can check that $F(z,x)$ is H-admissible, the coefficient extraction admits the classical saddle–point estimate

$$[t^n]F(t,x) \sim \frac{F(\zeta_x(n),x)}{\zeta_x(n)^n\sqrt{2\pi h''(\zeta_x(n),x)}}.$$

So one obtains

$$p_n(x) \sim \frac{F(\zeta_x(n),x)}{F(\zeta_1(n),1)}\left(\frac{\zeta_1(n)}{\zeta_x(n)}\right)^n K(x)(1+o(1)), \qquad K(x) = \frac{\zeta_1(n)}{\zeta_x(n)}\sqrt{\frac{h''(\zeta_1(n),1)}{h''(\zeta_x(n),x)}}.$$

Now, I want to apply the Generalized quasi-powers, we need $p_n = \exp(h_n)$, so

$$h_n(x) = \log p_n(x) = h(\zeta_x(n),x) - h(\zeta_1(n),1) + n\log\frac{\zeta_1(n)}{\zeta_x(n)} + \log K(x) + o(1).$$

Checking carefully, on has

$$\frac{h_n'''(1)}{\left(h_n'(1)+h_n''(1)\right)^{3/2}} \to 0 \quad (n\to\infty).$$

So the Generalized quasi-powers theorem implies that the random variable associated X_n converges in distribution to a normal law with mean $h_n'(1)$ and variance $h_n'(1)+h_n''(1)$.

..... COMPUTATION TO BE CHECK CAREFULLY AND CONTINUED... SOME CONDITIONS REQUIRED!!!

2.6 Case 3A: $b_1, a_2, b_1+b_2 \neq 0$

$$F(t,x) = \sum_{n,k} T(n,k)x^k\frac{t^n}{n!} = (1-(b_1+b_2)xt)^{-\frac{b_0+b_1+b_2}{b_1+b_2}}$$
$$\times\left(1-\frac{a_2}{b_1x}\left[1-(1-(b_1+b_2)xt)^{\frac{b_1}{b_1+b_2}}\right]\right)^{-\frac{a_0+a_2}{a_2}}.$$

Note that the singularity at $x=0$ is removable. We want to use Algebraic singularity schema, meanly we try to write

$$F(t,x) = A(t,x) + B(t,x)C(t,x)^{-\alpha}. \tag{21}$$

$A=0, B(t,x) = (1-(b_1+b_2)xt)^{-\frac{b_0+b_1+b_2}{b_1+b_2}}$ and $C(t,x) = 1-\frac{a_2}{b_1x}\left[1-(1-(b_1+b_2)xt)^{\frac{b_1}{b_1+b_2}}\right]$. Algebraic perturbation is verified for good choice of $r < \frac{1}{b_1+b_2}$. Under some conditions on the sign a_i, b_i, non-degeneracy is also verified... to write in details...

$$\rho(x) = \frac{1}{(b_1+b_2)x}\Big(1 - \big(1 - \frac{b_1}{a_2}x\big)^{\frac{b_1+b_2}{b_1}}\Big),$$

whose the singularity at $x = 0$ is removable.

..... AFTER computation (TO TAP LATER),
We see that

$$\mathfrak{v}\left(\frac{\rho(1)}{\rho(u)}\right) = -\frac{\rho''(1)}{\rho(1)} - \frac{\rho'(1)}{\rho(1)} + \left(\frac{\rho'(1)}{\rho(1)}\right)^2 \neq 0.$$

So the random variable X_n of probability generating function

$$p_n(x) = \mathbb{E}(x^{X_n}) = \frac{[t^n]F(t,x)}{[t^n]F(t,1)}$$

converges in distribution to a Gaussian with speed of convergence that is $O(n^{-\frac{1}{2}})$. The mean μ_n and the standard deviation σ_n^2 are asymptotically linear in n. More precisely $\mu_n = \mathfrak{m}\big(\frac{\rho(1)}{\rho(x)}\big)n$ and and $\sigma_n^2 = \mathfrak{v}\left(\frac{\rho(1)}{\rho(u)}\right)n$. Noting $\mathfrak{m}\big(\frac{\rho(1)}{\rho(x)}\big) = -\frac{\rho'(1)}{\rho(1)}$, we have explicitly

$$\mu_n = \left(1 - \tfrac{b_1+b_2}{a_2} \cdot \frac{\left(\frac{a_2-b_1}{a_2}\right)^{\frac{b_1+b_2}{b_1}-1}}{1-\left(\frac{a_2-b_1}{a_2}\right)^{\frac{b_1+b_2}{b_1}}}\right)n,\ \sigma_n^2 = \left[\tfrac{(b_1+b_2)b_2}{a_2^2} \cdot \frac{\left(\frac{a_2-b_1}{a_2}\right)^{\frac{b_1+b_2}{b_1}-2}}{1-\left(\frac{a_2-b_1}{a_2}\right)^{\frac{b_1+b_2}{b_1}}} + \left(\tfrac{b_1+b_2}{a_2} \cdot \frac{\left(\frac{a_2-b_1}{a_2}\right)^{\frac{b_1+b_2}{b_1}-1}}{1-\left(\frac{a_2-b_1}{a_2}\right)^{\frac{b_1+b_2}{b_1}}} - 1\right)^2\right]n$$

... CONDITIONs ON PARAMETERS WILL BE VERIFIED// CHECKED WITH THE COMPUTATIONS...

References

[1] D. Dumont, "William Chen grammars and derivations in trees and arborescences," *Séminaire Lotharingien de Combinatoire*, vol. 37, pp. B37a–21, 1996.

[2] P. Flajolet and R. Sedgewick, Analytic Combinatorics, *Cambridge University Press*, 2009. Available: https://doi.org/10.1017/CBO9780511801655

[3] H.-K. Hwang, H.-H. Chern, and G.-H. Duh, An asymptotic distribution theory for Eulerian recurrences with applications, *Advances in Applied Mathematics*, 112, Article 101960, 2020.

[4] R. L. Graham, D. E. Knuth, and O. Patashnik, *Concrete Mathematics: A Foundation for Computer Science*, 2nd ed., Addison-Wesley, Reading, MA, 1994.

[5] E. Neuwirth, Recursively defined combinatorial functions: extending Galton's board, *Discrete Mathematics*, 239(1), 33–51, 2001. doi:10.1016/S0012-365X(00)00373-3.

[6] M. Z. Spivey, On Solutions to a General Combinatorial Recurrence, *Journal of Integer Sequences*, 14, Article 11.9.7, 2011. Available: https://cs.uwaterloo.ca/journals/JIS/VOL14/Spivey/spivey31.pdf

[7] H. S. Wilf, The method of characteristics, and "problem 89" of Graham, Knuth and Patashnik, arXiv:math/0406620, 2004. https://arxiv.org/abs/math/0406620

[8] P. Hitczenko, On limiting distributions of Graham–Knuth–Patashnik recurrences, *Advances in Applied Mathematics*, 2020. (Note: verify volume and page numbers.)

[9] B. Randrianirina, *Espèces mixtes et système d'équations différentielles*. Annales des Sciences Mathématiques du Québec, vol. 24, no. 2, pp. 179–211, 2000, available online at `https://www.labmath.uqam.ca/annales/indexgeneral.html`.

[10] C. B. Corcino, R. B. Corcino, and N. Acala, Asymptotic Estimates for r-Whitney Numbers of the Second Kind, *Journal of Applied Mathematics*, vol. 2014, Article ID 354053, 2014. Available: https://doi.org/10.1155/2014/354053

[11] C. B. Corcino and R. B. Corcino, Asymptotic Estimates for Second Kind Generalized Stirling Numbers, *Journal of Applied Mathematics*, vol. 2013, Article ID 918513, 2013. Available: https://doi.org/10.1155/2013/918513